

ECOFLEET AI: INTELLIGENT ELECTRIC VEHICLE FLEET MONITORING AND ROUTE OPTIMIZATION PLATFORM

CSA1016 SOFTWARE ENGINEERING

SCENARIO-BASED REQUIREMENT ANALYSIS

NAME : KENISHA JUDITH J

REG NO : 192572071

DEPARTMENT : CSE (AI)

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1. INTRODUCTION

1.1 Overview

The rapid growth of electric vehicles (EVs) has transformed the transportation industry by providing a cleaner and more sustainable alternative to conventional fuel-powered vehicles. Logistics companies, delivery services, public transportation agencies, and fleet operators are increasingly adopting electric vehicles to reduce carbon emissions, improve fuel efficiency, and comply with environmental regulations. However, managing an electric vehicle fleet involves more than simply tracking vehicle locations. Fleet operators must continuously monitor battery health, charging schedules, vehicle performance, maintenance requirements, and route efficiency to ensure smooth operations.

The **EcoFleet AI: Intelligent Electric Vehicle Fleet Monitoring and Route Optimization Platform** is an Artificial Intelligence-based fleet management system designed to address these challenges. The platform integrates IoT sensors, GPS technology, cloud computing, and machine learning algorithms to monitor fleet performance in real time. It provides predictive analytics for battery health, recommends energy-efficient routes, optimizes charging schedules, and generates analytical reports to improve operational efficiency.

The system enables fleet managers to make informed decisions using real-time dashboards and predictive insights. By reducing unnecessary charging stops, preventing unexpected battery failures, and optimizing vehicle utilization, EcoFleet AI contributes to cost reduction, improved productivity, and sustainable transportation.

1.2 Purpose

The purpose of this Software Requirement Specification (SRS) is to define the functional and non-functional requirements of the EcoFleet AI platform. This document serves as a reference for developers, testers, project managers, and stakeholders throughout the software development lifecycle.

The SRS aims to:

- Clearly define the objectives of the proposed system.
- Describe the system functionality and operational scope.
- Identify all stakeholders and their responsibilities.
- Specify system requirements and constraints.
- Provide a foundation for system design, implementation, testing, and maintenance.

1.3 Intended Audience

This document is intended for:

- Project Manager
- Software Developers
- System Designers
- Database Administrators
- Quality Assurance Engineers
- Fleet Managers
- Maintenance Teams
- End Users
- Academic Evaluators

1.4 Document Organization

This document is organized into multiple sections covering the introduction, problem description, system scope, stakeholder analysis, requirements specification, system features, assumptions, constraints, validation, and conclusion.

2. INDUSTRY CONTEXT

The transportation sector is one of the fastest-growing industries adopting electric vehicles to support sustainable mobility. Governments worldwide are promoting electric mobility through subsidies, stricter emission regulations, and investments in charging infrastructure. Fleet operators in logistics, public transportation, and e-commerce increasingly rely on electric vehicles to reduce fuel costs and environmental impact.

Although electric vehicles offer significant advantages, managing an EV fleet presents several operational challenges. Fleet managers must monitor battery health, optimize charging schedules, minimize energy consumption, and ensure that vehicles remain available for service. Traditional fleet management systems mainly focus on GPS tracking and lack intelligent capabilities for predictive maintenance and energy optimization.

Recent advancements in Artificial Intelligence, Machine Learning, Internet of Things (IoT), GPS technology, cloud computing, and big data analytics have enabled the development of intelligent fleet management systems. These technologies allow continuous monitoring of vehicle health, prediction of battery degradation, optimization of charging schedules, and recommendation of energy-efficient routes based on traffic conditions and vehicle performance.

EcoFleet AI combines these modern technologies into a unified platform that supports intelligent decision-making. By integrating predictive analytics with real-time monitoring, the platform improves fleet productivity, reduces operational costs, extends battery lifespan, and contributes to sustainable transportation.

TECHNOLOGIES USED :

Technology	Purpose
Artificial Intelligence	Route optimization and predictive analytics
Machine Learning	Battery degradation prediction

IoT Sensors	Real-time vehicle monitoring
GPS	Vehicle tracking and navigation
Cloud Computing	Centralized data storage and processing
Big Data Analytics	Fleet performance analysis
Dashboard Analytics	Decision support and visualization

3. PROBLEM DESCRIPTION

Electric vehicle fleet operators face numerous operational challenges in managing battery performance, charging infrastructure, vehicle availability, maintenance schedules, and route planning simultaneously. Existing fleet management systems primarily provide vehicle tracking but do not offer intelligent features for predicting battery degradation, optimizing charging schedules, or minimizing energy consumption.

Poor route planning often results in unnecessary charging stops, increased operational expenses, delivery delays, and inefficient vehicle utilization. Battery failures and unexpected maintenance further reduce fleet availability, affecting business productivity and customer satisfaction. Additionally, fleet managers require real-time visibility into charging station availability, vehicle health, and energy consumption to make informed operational decisions.

As electric vehicle adoption continues to grow, there is a significant need for an intelligent fleet management platform capable of continuously monitoring vehicle performance, predicting maintenance requirements, optimizing charging operations, and recommending energy-efficient routes.

The proposed EcoFleet AI platform addresses these challenges by integrating Artificial Intelligence, Machine Learning, IoT sensors, GPS technology, and cloud computing into a unified intelligent system. The platform enables real-time monitoring, predictive analytics, automated decision support, and comprehensive reporting to improve fleet productivity while reducing operational costs and environmental impact.

Existing System

The existing fleet management systems generally provide:

- GPS-based vehicle tracking
- Basic fleet monitoring
- Manual charging management
- Limited maintenance scheduling
- Static route planning
- Manual report generation

Limitations of Existing System

- No battery health prediction
- No AI-based maintenance forecasting
- Poor charging schedule optimization
- High operational costs
- Increased energy consumption
- Limited decision support
- Lack of real-time analytics
- No intelligent route optimization

4. OBJECTIVES

The primary objective of the **EcoFleet AI: Intelligent Electric Vehicle Fleet Monitoring and Route Optimization Platform** is to develop an intelligent system that enhances the efficiency, sustainability, and reliability of electric vehicle fleet operations. The platform integrates Artificial Intelligence (AI), Internet of Things (IoT), GPS technology, and Machine Learning to provide real-time monitoring, predictive maintenance, and route optimization.

4.1 Objectives

1. Develop an AI-powered platform to monitor electric vehicle fleet performance in real time.
2. Continuously monitor battery health, energy consumption, and vehicle status using IoT sensors.
3. Predict battery degradation and maintenance requirements using machine learning algorithms.
4. Optimize charging schedules based on battery condition, electricity demand, and charging station availability.
5. Recommend energy-efficient routes by considering GPS location, traffic conditions, and battery level.
6. Provide a real-time dashboard for fleet managers to monitor all vehicle operations.
7. Generate analytical reports on fleet utilization, energy efficiency, and operational performance.
8. Improve fleet productivity while reducing operational costs and supporting sustainable transportation

5. SCOPE OF THE SYSTEM

The EcoFleet AI platform is designed to provide an intelligent solution for managing electric vehicle fleets in logistics, transportation, and delivery industries. The system continuously collects real-time data from IoT sensors installed in electric vehicles and processes this information using Artificial Intelligence and Machine Learning techniques.

The platform enables fleet managers to monitor battery performance, track vehicle location, predict maintenance requirements, optimize charging schedules, and recommend energy-efficient routes. It also provides dashboards and analytical reports to support decision-making and improve overall fleet efficiency.

The system aims to reduce charging costs, extend battery life, minimize vehicle downtime, and improve fleet utilization. It can be deployed for small, medium, and large-scale fleet operations and is scalable to support future smart transportation technologies.

6. EXISTING SYSTEM

Existing electric vehicle fleet management systems mainly focus on vehicle tracking and basic fleet monitoring. Most systems provide GPS-based location tracking and simple maintenance reminders. However, they lack intelligent features such as predictive battery analysis, AI-based maintenance prediction, and optimized charging schedules.

These limitations reduce operational efficiency and increase energy consumption, charging costs, and vehicle downtime.

Limitations of Existing System

- Limited battery health monitoring.
- No prediction of battery degradation.
- Manual charging schedule management.

- No AI-based route optimization.
- High operational and maintenance costs.
- Limited real-time analytics.
- Poor fleet utilization.
- Lack of intelligent decision support.

7. PROPOSED SYSTEM

The proposed **EcoFleet AI** platform integrates Artificial Intelligence, IoT, GPS, Cloud Computing, and Machine Learning to create an intelligent fleet management solution.

The platform continuously monitors battery health, vehicle status, energy consumption, and charging station availability. AI algorithms analyze historical and real-time data to predict battery degradation and maintenance requirements before failures occur.

The system also recommends energy-efficient routes based on traffic conditions, battery level, and charging station availability. Fleet managers can monitor all vehicles through a centralized dashboard and generate analytical reports to improve operational performance.

Advantages of the Proposed System

- Intelligent battery health monitoring.
- Predictive maintenance using AI.
- Optimized charging schedules.
- Energy-efficient route recommendations.
- Reduced operational costs.
- Improved fleet utilization.
- Real-time monitoring and alerts.
- Enhanced decision-making through dashboards and reports.

8. STAKEHOLDER ANALYSIS

Stakeholder	Role and Responsibility
Fleet Manager	Monitors fleet operations, assigns vehicles, reviews reports, and makes operational decisions.
Driver	Operates assigned electric vehicles, follows optimized routes, and receives notifications.
Administrator	Manages users, vehicles, system settings, and security policies.
Maintenance Technician	Performs vehicle maintenance based on AI-generated maintenance alerts.
Charging Station Operator	Provides charging infrastructure information and charging availability.
AI Prediction Engine	Predicts battery degradation, charging requirements, and maintenance schedules.
GPS Service	Provides real-time vehicle location and traffic information.
IoT Sensors	Continuously collect battery, temperature, speed, and energy consumption data.
Cloud Server	Stores fleet data, processes analytics, and provides system accessibility.

9. FUNCTIONAL REQUIREMENTS

Functional requirements describe the specific services and operations that the EcoFleet AI system must perform to meet user needs.

Requirement ID	Functional Requirement
FR1	The system shall allow users to log in securely.
FR2	The system shall allow administrators to manage user accounts.
FR3	The system shall register electric vehicles into the fleet.
FR4	The system shall monitor the real-time location of all vehicles.
FR5	The system shall monitor battery health continuously using IoT sensors.
FR6	The system shall display vehicle status (Active, Charging, Maintenance, Offline).
FR7	The system shall predict battery degradation using AI models.
FR8	The system shall predict maintenance requirements.
FR9	The system shall recommend energy-efficient routes.
FR10	The system shall optimize charging schedules.
FR11	The system shall display nearby charging stations.
FR12	The system shall track energy consumption.
FR13	The system shall generate maintenance alerts.
FR14	The system shall generate battery health alerts.
FR15	The system shall send notifications to fleet managers.
FR16	The system shall assign vehicles to drivers.

FR17	The system shall manage driver information.
FR18	The system shall display traffic information.
FR19	The system shall generate fleet performance reports.
FR20	The system shall generate energy consumption reports.
FR21	The system shall generate battery performance reports.
FR22	The system shall maintain vehicle history records.
FR23	The system shall maintain charging history.
FR24	The system shall allow administrators to configure system settings.
FR25	The system shall provide a centralized fleet dashboard.

10. NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements specify the quality attributes and operational characteristics of the EcoFleet AI platform.

Performance Requirements

- The system shall respond to user requests within 3 seconds.
- The dashboard shall update real-time data every 5 seconds.
- The platform shall support monitoring of at least 1,000 vehicles simultaneously.

Security Requirements

- User authentication shall be required.
- Passwords shall be encrypted.
- Communication shall use secure protocols (HTTPS).
- Access shall be role-based.

Reliability

- System availability shall be at least 99.9%.
- Automatic backup shall be performed daily.
- Data loss shall be minimized during failures.

Usability

- The interface shall be user-friendly.
- Dashboard navigation shall be simple and intuitive.
- Reports shall be easy to understand.

Scalability

- The platform shall support future expansion.
- New vehicles and charging stations shall be added without affecting performance.

Maintainability

- The system shall support easy software updates.
- System logs shall be maintained for troubleshooting.

Availability

- The system shall operate 24×7.
- Real-time monitoring shall be continuously available.

11. ACTORS AND USE CASES

11.1 Primary Actors :

Actor	Description
Administrator	Manages the entire system and user accounts.
Fleet Manager	Monitors fleet operations and generates reports.
Driver	Operates the assigned electric vehicle.
Maintenance Technician	Performs maintenance based on AI alerts.

11.2 Secondary Actors :

Actor	Description
GPS Service	Provides real-time location and traffic information.
IoT Sensors	Collect battery and vehicle performance data.
Charging Station	Provides charging availability information.
AI Prediction Engine	Predicts battery degradation and maintenance schedules.
Cloud Database	Stores system data and analytics.

11.3 Major Use Cases

- Login
- Manage Users
- Register Vehicle
- Monitor Fleet
- Track Vehicle Location
- Monitor Battery Health
- Predict Battery Degradation
- Predict Maintenance
- Optimize Charging Schedule
- Recommend Best Route
- Assign Driver
- Generate Reports
- Receive Notifications
- View Dashboard
- Logout

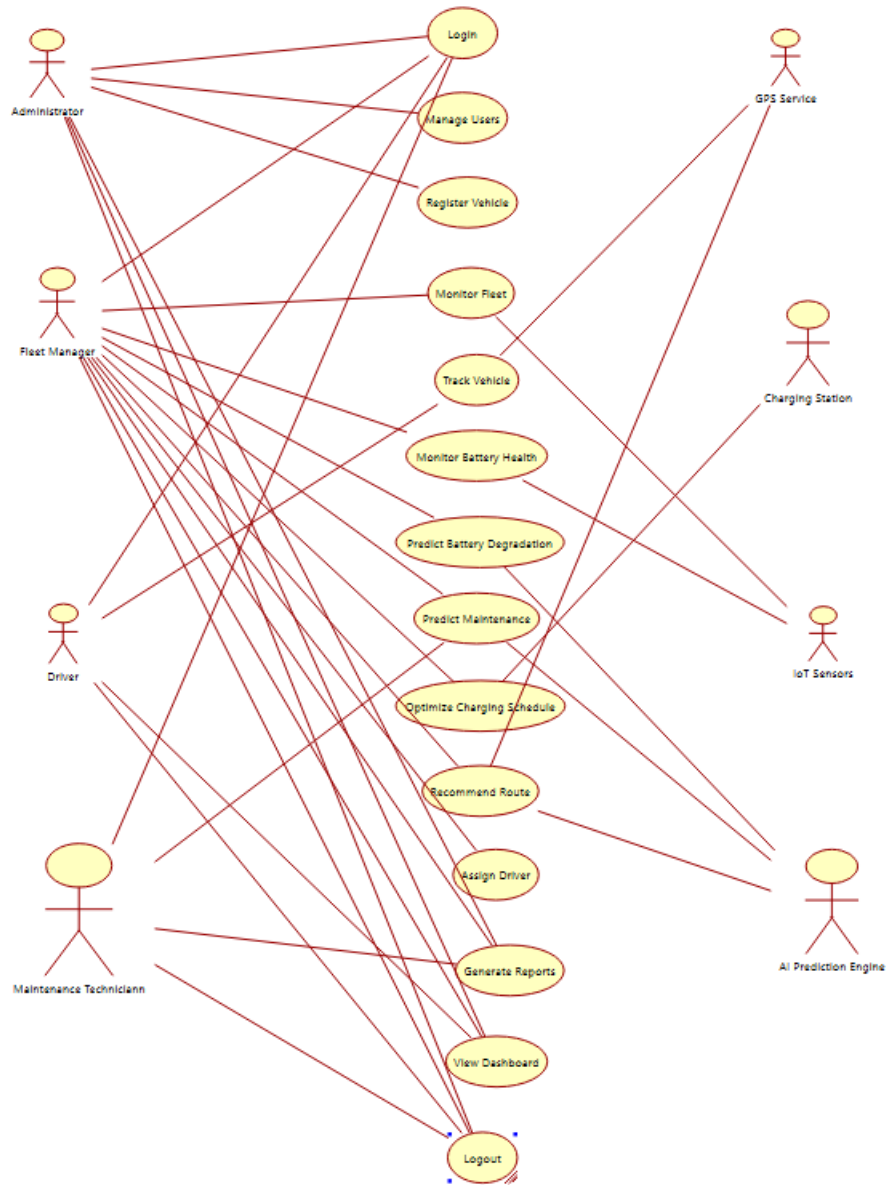


Figure 11.1: Use Case Diagram of EcoFleet AI System

12. USE CASE DESCRIPTIONS

Use Case 1: User Login:

Item	Description
Use Case ID	UC-01
UseCase Name	User Login
Primary Actor	Administrator / Fleet Manager / Driver
Description	Authenticates the user before accessing the system.
Preconditions	User account must exist.
Main Flow	Enter username → Enter password → Validate credentials → Access dashboard.
Postconditions	User successfully logs into the system.

Use Case 2: Monitor Battery Health:

Item	Description
Use Case ID	UC-02
Use Case Name	Monitor Battery Health
Primary Actor	Fleet Manager
Description	Displays battery status, charge level, and health.
Preconditions	Vehicle must be connected to IoT sensors.
Main Flow	Select vehicle → Retrieve battery data → Display battery information.
Postconditions	Fleet manager views current battery status.

Use Case 3: Optimize Charging Schedule :

Item	Description
Use Case ID	UC-03
Use Case Name	Optimize Charging Schedule
Primary Actor	Fleet Manager
Description	AI recommends the best charging schedule.
Preconditions	Battery and charging station data must be available.
Main Flow	Analyze battery level → Check charging station availability → Recommend charging time.
Postconditions	Charging schedule is optimized.

Use Case 4: Generate Fleet Reports:

Item	Description
Use Case ID	UC-04
Use Case Name	Generate Reports
Primary Actor	Fleet Manager
Description	Generates reports on fleet performance and energy usage.
Preconditions	Fleet data must be available.
Main Flow	Select report type → Generate report → Display and export report.
Postconditions	Reports are available for analysis.

13. DATA REQUIREMENTS

The EcoFleet AI platform relies on accurate and real-time data to monitor electric vehicle fleet operations, optimize routes, predict maintenance requirements, and generate analytical reports. The following data are required for the efficient functioning of the system.

13.1 Vehicle Data

- Vehicle ID
- Vehicle Model
- Registration Number
- Battery Capacity
- Current Battery Level
- Vehicle Status (Active, Charging, Maintenance)
- Energy Consumption
- Odometer Reading

13.2 Driver Data

- Driver ID
- Driver Name
- Contact Number
- Driving License Number
- Assigned Vehicle
- Duty Schedule

13.3 Battery Data

- Battery Health Percentage
- Charging Cycles
- Temperature
- Voltage

- State of Charge (SOC)
- Estimated Remaining Range

13.4 Charging Station Data

- Charging Station ID
- Location
- Number of Charging Ports
- Availability Status
- Charging Cost
- Charging Speed

13.5 GPS Data

- Vehicle Latitude
- Vehicle Longitude
- Route Information
- Distance Covered
- Estimated Time of Arrival (ETA)
- Traffic Conditions

13.6 System Reports

- Fleet Utilization Report
- Battery Performance Report
- Energy Consumption Report
- Maintenance Report
- Route Optimization Report

14. EXTERNAL INTERFACE REQUIREMENTS

The EcoFleet AI platform interacts with users, hardware devices, software services, and communication networks to provide intelligent fleet management.

14.1 User Interface

The system provides a user-friendly dashboard that enables fleet managers, administrators, and maintenance personnel to access real-time fleet information.

Features include:

- Secure Login Screen
- Dashboard
- Vehicle Monitoring Screen
- Battery Health Dashboard
- Charging Schedule Interface
- Route Recommendation Screen
- Report Generation Screen

14.2 Hardware Interface

The platform interacts with:

- IoT Sensors
- GPS Devices
- Electric Vehicle Battery Management System (BMS)
- Charging Stations
- Cloud Servers
- Desktop and Mobile Devices

14.3 Software Interface

The system integrates with:

- AI Prediction Engine

- Machine Learning Models
- GPS Mapping Services
- Cloud Database
- Notification Service
- Reporting Module

14.4 Communication Interface

Communication is performed through:

- Wi-Fi
- 4G/5G Networks
- HTTPS Protocol
- IoT Communication Protocols
- Cloud APIs

15. SYSTEM FEATURES

The EcoFleet AI platform offers several intelligent features that improve fleet management efficiency.

Key Features

- Real-Time Vehicle Tracking
- Battery Health Monitoring
- AI-Based Battery Degradation Prediction
- Predictive Maintenance
- Smart Charging Schedule Optimization
- Energy-Efficient Route Recommendation
- Driver Management
- Fleet Monitoring Dashboard

- Real-Time Notifications
- Analytical Report Generation

15.1 System Workflow

The workflow of the EcoFleet AI platform begins when the fleet manager logs into the system. The platform continuously receives real-time information from IoT sensors, GPS devices, and charging stations. AI algorithms analyze the collected data to predict battery degradation, recommend charging schedules, and identify maintenance requirements. Based on traffic conditions and battery status, the system recommends the most energy-efficient routes. Finally, the platform generates reports and displays real-time analytics through the dashboard.

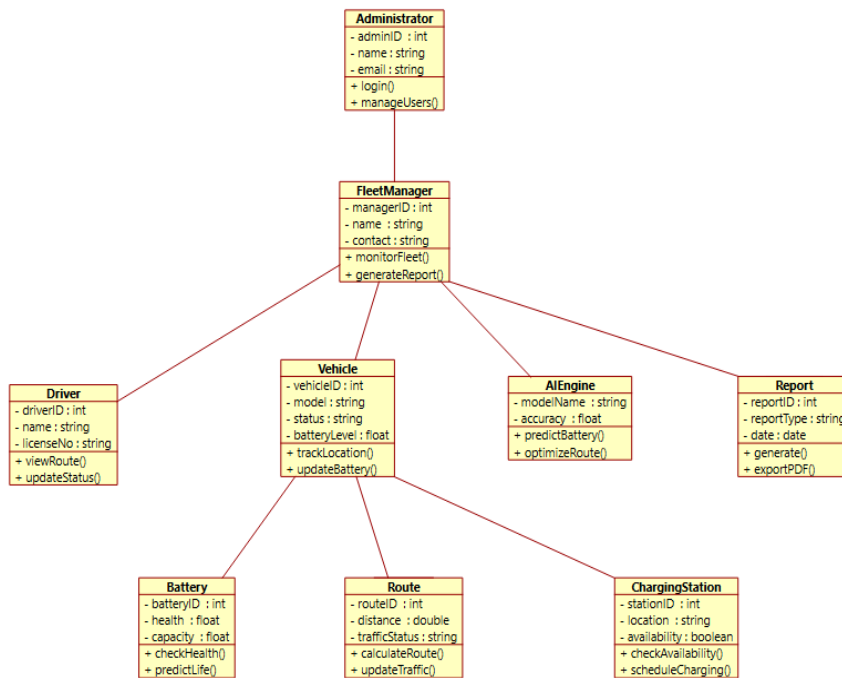


Figure 15.1: Activity Diagram of EcoFleet AI System

15.2 System Structure

The EcoFleet AI platform consists of several interconnected modules, including Vehicle Management, Driver Management, Battery Monitoring, Charging Management, AI Prediction Engine, Route Optimization, Dashboard, and Reporting Module. These modules interact with one another to provide intelligent fleet management.

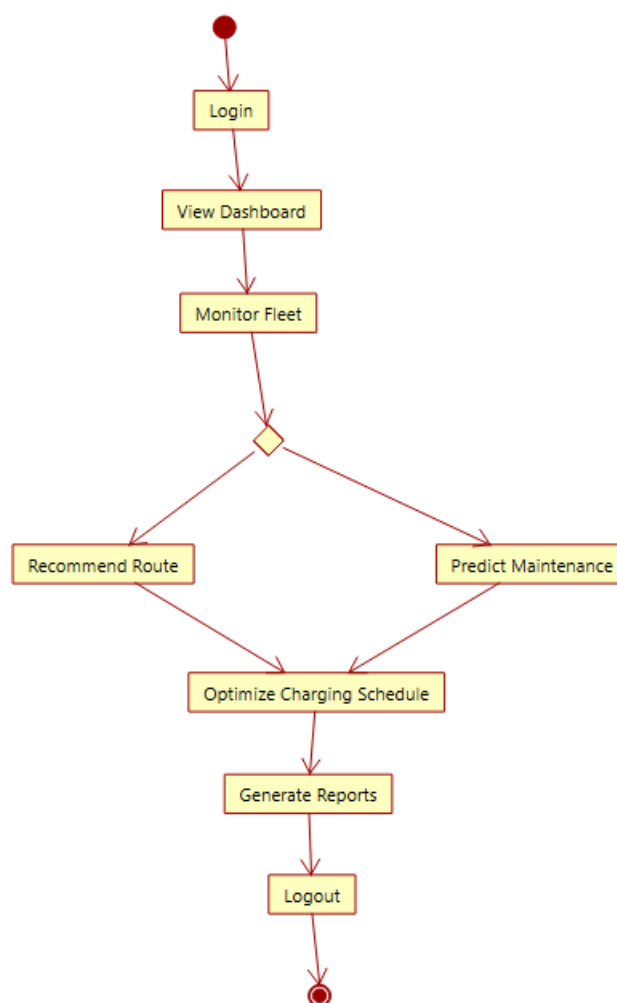


Figure 15.2: Class Diagram of EcoFleet AI System

16. Assumptions

The following assumptions have been made during the development of the EcoFleet AI platform:

1. All electric vehicles are equipped with IoT sensors.
2. GPS devices provide accurate location data.
3. Charging stations continuously update their availability.
4. Internet connectivity is available during fleet operations.
5. AI prediction models are trained using sufficient historical data.
6. Fleet managers have authorized access to the system.
7. Drivers use the recommended routes.
8. Cloud servers are available for data storage and processing.
9. Battery sensors provide accurate measurements.
10. Users possess basic computer knowledge.

17. Constraints

The following constraints may affect system performance:

Technical Constraints

- Dependence on stable internet connectivity.
- Sensor failures may affect monitoring accuracy.
- GPS inaccuracies in certain locations.

Operational Constraints

- Limited charging station availability.
- Battery charging time.
- Driver compliance with recommended routes.

Economic Constraints

- High installation cost of IoT devices.
- Cloud infrastructure expenses.

- AI model development cost.

Legal Constraints

- Compliance with transportation regulations.
- Protection of user and fleet data.
- Adherence to data privacy laws.

Time Constraints

- Real-time data processing requirements.
- Immediate alert generation during emergencies.

18. Requirement Validation

Requirement validation ensures that the system requirements accurately represent stakeholder needs and can be implemented successfully.

18.1 Completeness

All functional and non-functional requirements required for intelligent EV fleet management have been identified.

18.2 Consistency

The requirements are consistent with the project objectives and do not conflict with one another.

18.3 Correctness

The requirements support AI-based monitoring, predictive maintenance, charging optimization, and route recommendation.

18.4 Feasibility

The proposed system can be implemented using existing technologies such as AI, IoT, GPS, cloud computing, and machine learning.

18.5 Suggested Improvements

- Integration with renewable energy charging stations.
- Mobile application support.
- Voice-assisted fleet management.
- Advanced battery analytics.

- Carbon emission tracking.
- Integration with smart city infrastructure.

19. CONCLUSION

The **EcoFleet AI: Intelligent Electric Vehicle Fleet Monitoring and Route Optimization Platform** provides an intelligent solution for managing electric vehicle fleets through the integration of Artificial Intelligence, Internet of Things, GPS technology, and cloud computing. The platform continuously monitors battery health, predicts maintenance requirements, optimizes charging schedules, and recommends energy-efficient routes to improve fleet productivity.

The proposed system supports sustainable transportation by reducing operational costs, extending battery life, minimizing downtime, and improving vehicle utilization. Real-time dashboards and analytical reports enable fleet managers to make informed decisions, resulting in enhanced operational efficiency and reduced carbon emissions.

The implementation of EcoFleet AI will contribute significantly to the development of smart transportation systems and sustainable mobility solutions.

Appendix A: Project Management Using Jira :

Jira was used to manage the development activities of the EcoFleet AI project. Tasks were organized using a Kanban board with the stages To Do, In Progress, In Review, and Done. Modules such as Fleet Monitoring, Battery Health Monitoring, Route Optimization, Charging Schedule, Dashboard, and Report Generation were tracked throughout the development process. This approach improved task organization, progress monitoring, and overall project management.

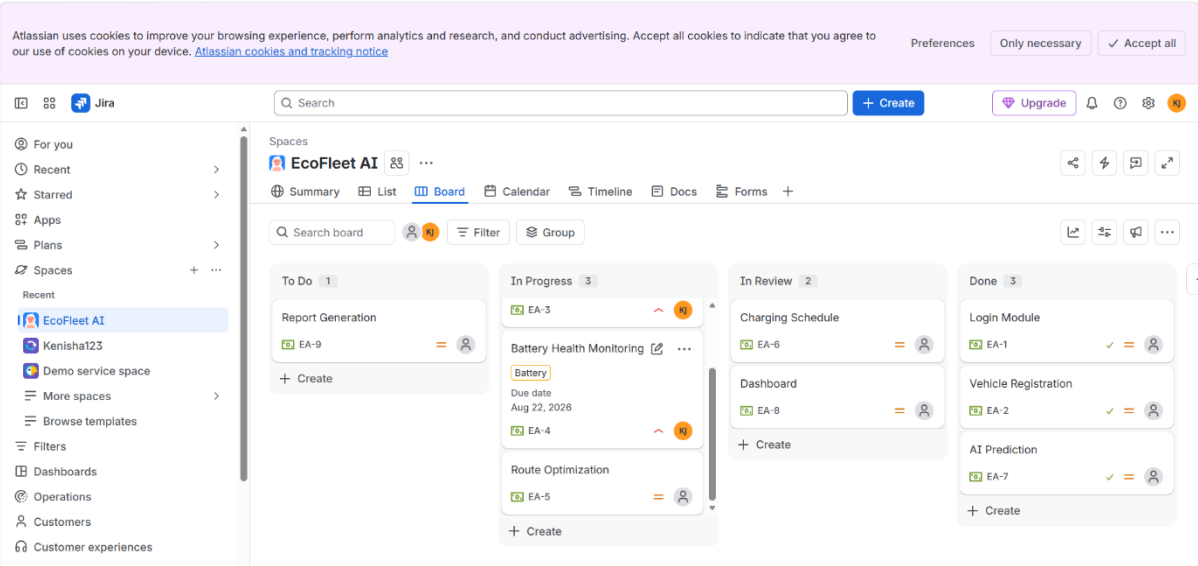


Figure A.1: Jira Kanban Board

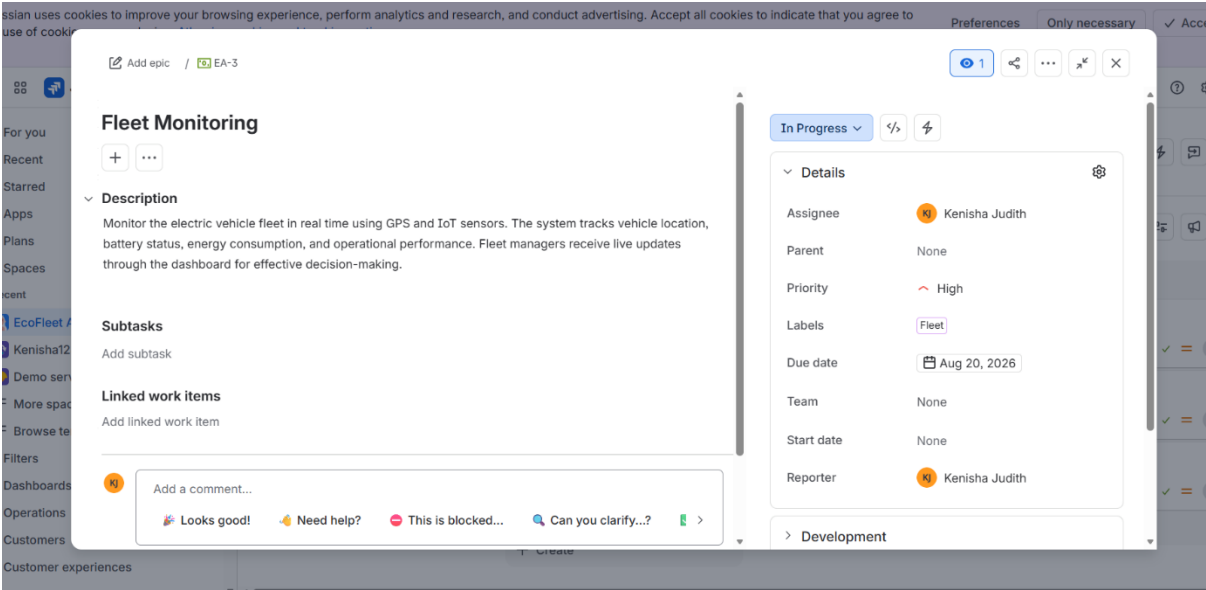


Figure A.2: Fleet Monitoring Task Details

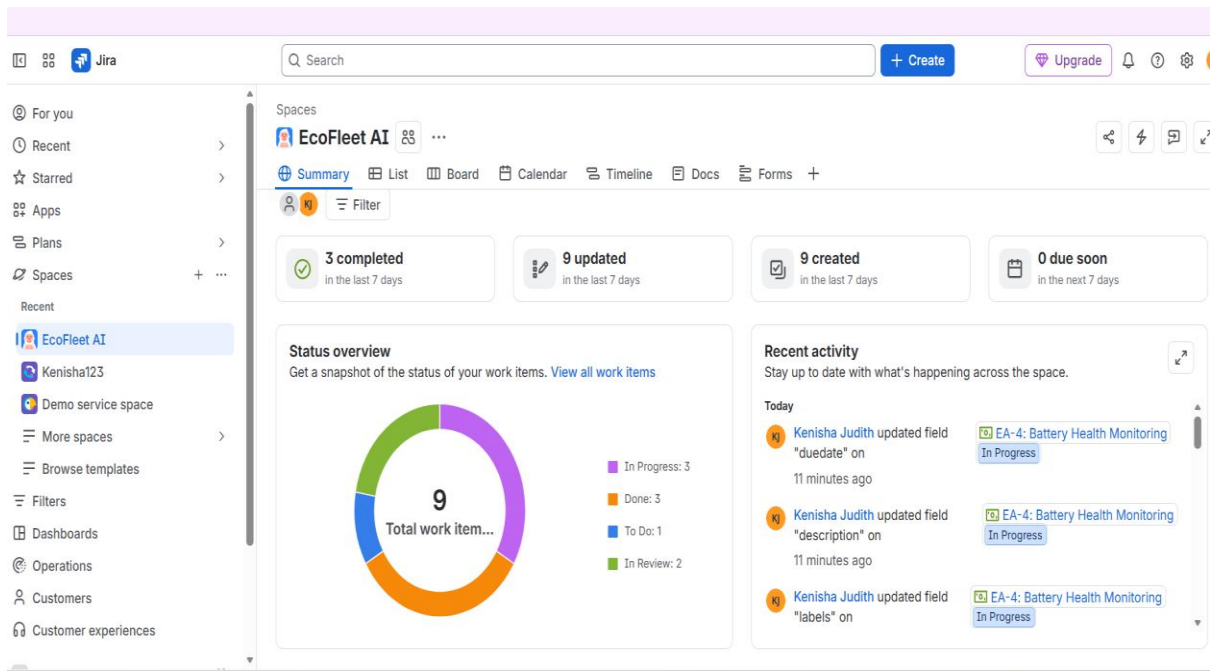


Figure A.3: Jira Project Summary